

BASE experiment takes a big step towards portable antimatter

The experiment successfully transported a box filled with unbonded protons across CERN's main site, thus demonstrating that the same feat could later be possible for antiprotons



The BASE-STEP transportable trap system, lifted by crane through the AD hall before being loaded onto a lorry. The team monitored all the parameters during transport. (Image: CERN)

Antimatter might sound like something out of science fiction, but at the CERN [Antiproton Decelerator](#) (AD), scientists produce and trap antiprotons every day. The [BASE experiment](#) can even contain them for more than a year—an impressive feat considering that antimatter and matter annihilate upon contact. The CERN AD hall is the only place in the world where scientists are able to store and study antiprotons. But this is something that scientist working on the BASE experiment hope to change one day with their subproject [BASE-STEP](#): an apparatus designed to store and transport antimatter.

Yesterday, the team of scientists and engineers took an important step towards this goal by transporting a cloud of 70 protons in a truck across CERN's main site. "If you can do it with protons, it will also work with antiprotons," said Christian Smorra, the leader of BASE-STEP. "The only difference is that you need a much better vacuum chamber for the antiprotons." This is the first time that loose particles have been transported in a reusable trap that scientists can then open in a new location and then transfer the contents into another experiment. The end goal is to create an antiproton-delivery service from CERN to experiments located at other laboratories.

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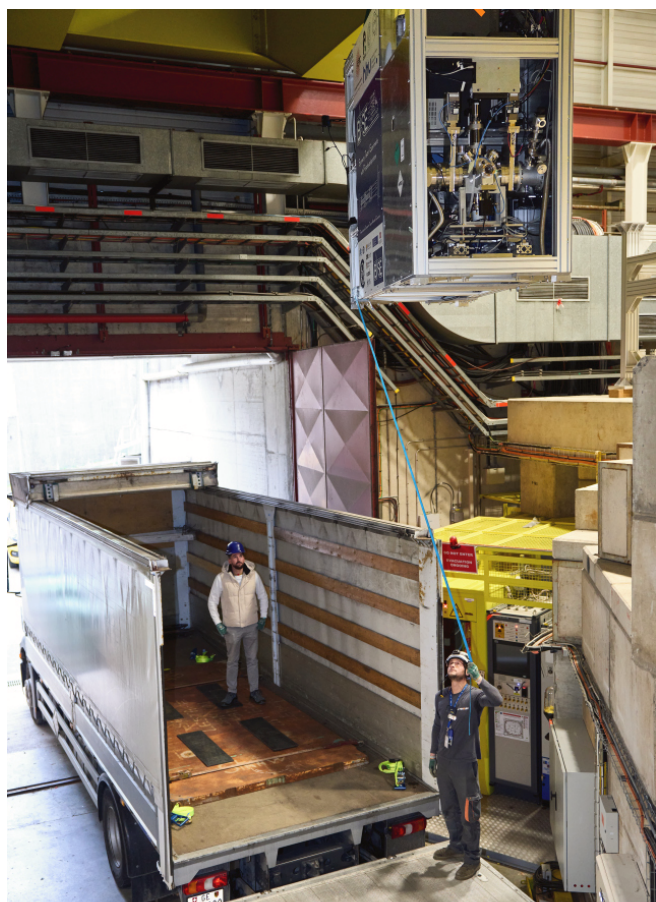
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Proactively tackling conflicts

Antimatter is a naturally occurring class of particles that is almost identical to ordinary matter except that the charges and magnetic properties are reversed. This has baffled scientists for decades because according to the laws of physics, the Big Bang should have produced equal amounts of matter and antimatter. These equal-but-opposite particles would have quickly annihilated with each other; leaving a simmering but empty Universe. Physicists suspect that there are hidden differences that can explain why matter survived and antimatter all but disappeared.

The BASE experiment aims to answer this question by precisely measuring the properties of antiprotons, such as their intrinsic magnetic moment, and then comparing these measurements with those taken with protons. However, the precision the experiment can achieve is limited by its location.



The transportable trap being carefully loaded in the truck before going for a road trip across CERN's main site. (Image: CERN)

“The accelerator equipment in the AD hall generates magnetic field fluctuations that limit how far we can push our precision measurements,” said BASE spokesperson Stefan

Ulmer. “If we want to get an even deeper understanding of the fundamental properties of antiprotons, we need to move out.”

This is where BASE-STEP comes in. The goal is to trap antiprotons and then transfer them to a facility where scientists can study them with a greater precision. To be able to do this, they need a device that is small enough to be loaded onto a truck and can resist to the bumps and vibrations that are inevitable during ground transport. The current apparatus — which includes a superconducting magnet, cryogenic cooling, power reserves, and a vacuum chamber that traps the particles using magnetic and electric fields — weighs 1000 kilograms and needs two cranes to be lifted out of the experimental hall and onto the truck. Even though it weighs a tonne, BASE-STEP is much more compact than any existing system used to study antimatter. For example, it has a footprint that is five times smaller than the original BASE experiment as it has to be narrow enough to fit through ordinary laboratory doors.

During the rehearsal, the scientists used trapped protons as a stand-in for antiprotons. Protons are a key ingredient of every atom, the simplest of which is hydrogen (one proton and one electron.) But storing protons as loose particles and then moving them onto a truck is a challenge because any tiny disturbance will draw the unbonded protons back into an atomic nucleus.

“When it’s transported by road, our trap system is exposed to acceleration and vibrations, and laboratory experiments are usually not designed for this”, Smorra said. “We needed to build a trap system that is robust enough to withstand these forces, and we have now put this to a real test for the first time.” However, Smorra noted that the biggest potential hurdle isn’t currently the bumpiness of the road but traffic jams. “If the transport takes too long, we will run out of helium at some point,” he said. Liquid helium keeps the trap’s superconducting magnet at a temperature below 8.2 Kelvin: its maximum operating temperature. If the drive takes too long, the magnetic field will be lost and the trapped particles will be released and vanish as soon as they touch ordinary matter.

“Eventually we want to be able to transport antimatter to our dedicated precision laboratories

at the Heinrich Heine University in Düsseldorf, which will allow us to study antimatter with at least 100-fold improved precision,” Smorra said. “In the longer term, we want to transport it to any laboratory in Europe. This means that we need to have a power generator on the truck. We are currently investigating this possibility.”

After this successful test, which included ample monitoring and data taking, the team plans to refine its procedure with the goal of transporting antimatter next year. “This is a totally new technology that will open the door for new possibilities of study, not only with antiprotons but also with other exotic particles, such as ultra-highly-charged ions,” Ulmer said.

Another experiment, PUMA, is preparing a transportable trap. Next year, it plans to transport antiprotons next year 600 metres from the AD hall to CERN’s ISOLDE facility in order to use it to study

the properties and structure of exotic atomic nuclei.



The BASE-STEP team celebrating the successful transport at the end of the operation. The green signal on the computer screen shows that the 70 loose protons are still "alive", maintained by the magnetic field in the trap. (Image: CERN)

Sarah Charley

Accelerator Report: Lead ions take the stage

As the LHC and SPS wrap up their 2024 proton run, teams are making final preparations for the lead-ion physics run starting on 4 November

On 30 October at 8 a.m., the SPS delivered its final protons of the year to the North Area, marking the shift to the lead-ion physics run.

To prepare for this transition, the BE departmental safety officer (DSO), with support from various experts, conducted this morning the SPS North Area lead-ion DSO test. This test involves verifying the additional safety measures required for lead-ion operations, ensuring that no protons can accidentally be directed to the North Area. The DSO test includes evaluating fault scenarios, confirming safety protocols and validating all the measures in place. Once it was completed, the lead-ion beam was extracted and directed towards the North Area, where beamline physicists are now adjusting the beam trajectories and optics to deliver high-quality lead-ion beam to the experiments.

Approximately 30 hours after the last proton beam to the North Area was stopped, a radiation survey will be conducted in the early afternoon of 1 November. During this survey, radiation protection specialists will measure radiation levels

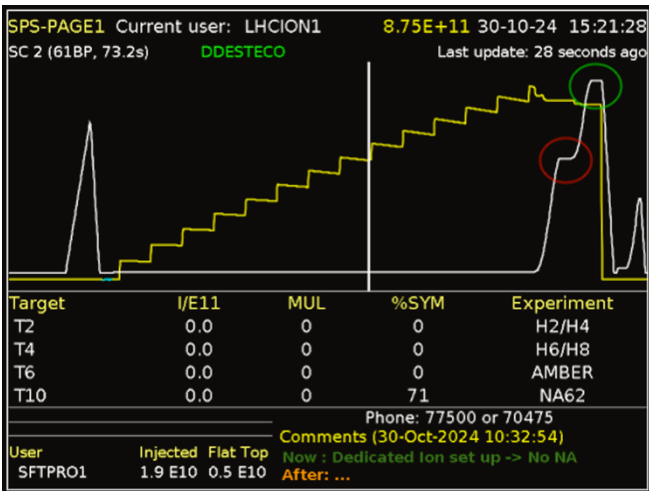
along the machines (Linac4, PSB, PS and SPS). Annual comparisons of these measurements help track radiation level changes and identify areas where levels may have increased or decreased. Machine experts are then consulted to understand the potential causes of any changes. If any anomalies are detected, they can be investigated further and corrected for future runs. This approach helps to keep radiation levels as low as reasonably possible, minimising downtime and ensuring safe conditions for machine interventions following a predefined cooldown period.

Over the weekend of 2 and 3 November, the final adjustments will be made to the SPS lead-ion beam production and North Area beamlines before starting the four-week lead-ion physics run in the North Area on 4 November, which will be concluded at 6 a.m. on 2 December.

The lead-ion beam from the PS is also used in the PS East Area by the [HEARTS](#) (High-Energy Accelerators for Radiation Testing and Shielding) project, which will start its experiments on 11

November and will also conclude at 6 a.m. on 2 December.

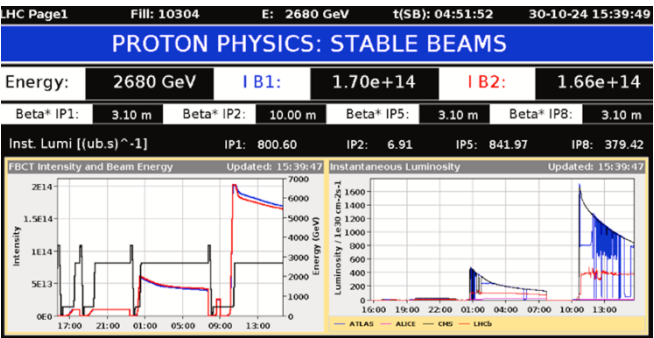
The lead-ion beam for the LHC is of a unique type and has been meticulously prepared over recent weeks. This involved five dedicated 10-hour sessions during which SPS proton physics was paused in order to allow expert teams to set up the LHC lead-ion beam in the SPS. This beam requires a series of 14 injections, each containing four lead-ion bunches from the PS, resulting in 56 bunches spaced by 100 ns. After an initial acceleration, a process called [slip-stacking](#) reduces the bunch spacing to 50 ns, allowing more bunches to be injected into the LHC. The beam then undergoes a final acceleration, reaching the final flat-top, after which it is extracted and injected into the LHC in either a clockwise or counterclockwise direction.



The SPS display showing the LHC lead-ion production cycle. In white is the energy and in yellow the number of lead ions, clearly showing the 14 injections. The intermediate flat-top (red circle) is used to perform the slip-stacking and is followed by the acceleration to the final flat-top (green circle) where the beam is extracted. (Image: CERN)

The initial set-up of lead ions in the LHC, using only a few bunches, took place over the weekend of 26 and 27 October. Currently, the LHC is colliding protons at 2.68 TeV – compared with the usual 6.8

TeV – for the proton–proton reference run, providing calibration data for the upcoming lead-ion collisions. Final adjustments for the lead-ion beam set-up in the LHC are scheduled for the coming weekend and should be concluded on 4 November, with lead-ion collisions for physics set to begin later that day. The LHC lead-ion run will conclude at 6 a.m. on 25 November, with the injector complex remaining operational for another week to provide beam to the experimental facilities.



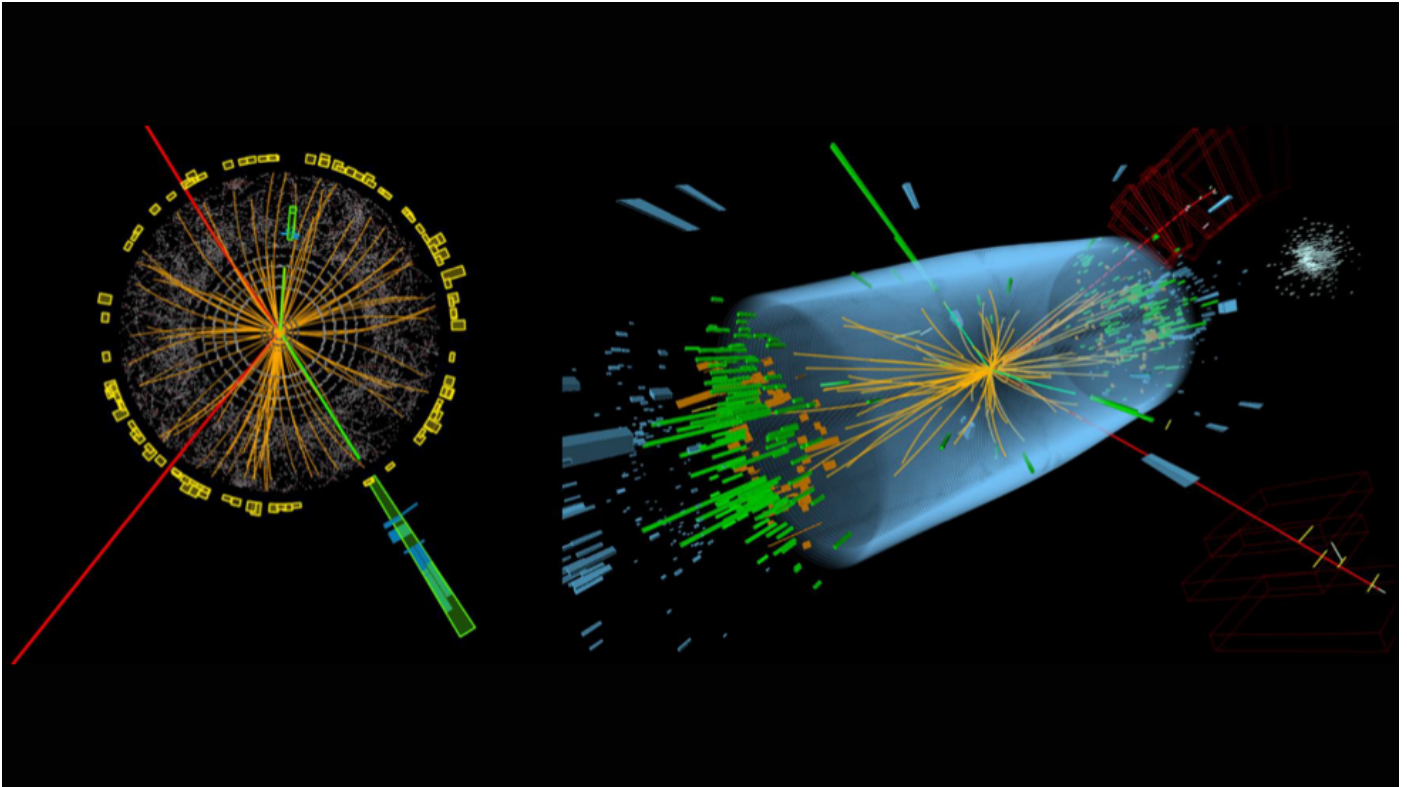
The LHC display showing the proton–proton reference runs. In the bottom left-hand graph in blue and red are the clockwise and counterclockwise proton beams. In black is the energy, with the right-hand scale indicating 2.68 TeV. The small spike at the end of the flat-top, which occurs after the beams are dumped and goes up to 3.5 TeV, is required to ensure that the magnetic conditions of the LHC machine are correct for the next beam to be injected. (Image: CERN)

As mentioned in my [previous report](#), additional enhancements to the lead-ion beam’s performance as well as efforts to reduce background, particularly for the ALICE experiment, are essential. Last weekend offered the first promising signs of these improvements, and there is much anticipation for the final outcomes with full beams. Next week will bring further answers, hopefully confirming these positive developments.

Rende Steerenberg

CERN70: Announcing the Higgs boson discovery

Part 20 of the [CERN70](#) feature series. Fabiola Gianotti, CERN Director-General, was spokesperson of the ATLAS experiment at the Large Hadron Collider from 2009 to 2013 and responsible for announcing the discovery of the Higgs boson, alongside the CMS spokesperson



Candidate Higgs boson events from collisions between protons in the LHC, in the ATLAS experiment (left) and in the CMS experiment (right). (Images: [ATLAS](#)/[CMS](#)/[CERN](#))

Ask any member of the particle physics community where they were on 4 July 2012 and they won't need to think too hard about it. The discovery of the [Higgs boson](#) was a milestone in the history of science, as evidenced by the media tsunami that it caused.

At 9.00 a.m. on 4 July 2012, Joe Incandela and Fabiola Gianotti, the spokespersons for the [CMS](#) and [ATLAS](#) experiments, took the floor one after the other in front of an excited audience to present the latest data from their experiments. No news had yet been released, but all of the experts were anticipating a spectacular announcement. The conference was broadcast in dozens of institutes around the world. In the United States, scientists woke up in the middle of the night to attend the event, on a bank holiday no less. At 10.40 a.m., thunderous applause erupted in CERN's Main Auditorium, Peter Higgs shed a tear and the Director-General, [Rolf Heuer](#), declared: "As a layman, I would say: now we have it!". The results were unequivocal: they revealed the presence of a particle whose properties matched those of the Higgs boson, a particle that had been predicted 48 years earlier.

[The story starts in the 1960s](#), when physicists began to realise that the electromagnetic and weak interactions (two of nature's four fundamental forces) might be described by the same mathematical structure. A major challenge in building this unified "electroweak" theory, which is now a pillar of the [Standard Model of particle physics](#), was that its equations demanded that the mediating particles of the interactions have zero mass, which would imply that they have an infinite range. But while the photon, the vector of the electromagnetic force, is indeed devoid of mass, this can't be the case for the carriers of the weak interaction, as this interaction has a very small range, on an atomic scale.



Belgian theorist, François Englert (left), and British theorist, Peter Higgs (right), in CERN's Main Auditorium, speaking after the announcement of the discovery on 4 July 2012. They were awarded the Nobel Prize in Physics the following year. (Image: CERN)

The solution to this problem lay in work published earlier by the Belgian theorists Robert Brout and François Englert and, independently, the British theorist Peter Higgs. Taking inspiration from extensive research work – in particular, work on superconductivity – they had [proposed a mechanism](#) that, when it was applied by Steven Weinberg to the electroweak theory, enabled the mediators of the weak force to gain mass while ensuring that the photon of electromagnetism did not. The mechanism had two profound implications: the existence of the [Z boson](#) (a further carrier of weak interactions, in addition to the [W boson](#)), and the existence of an invisible field that permeates the entire Universe, with which more familiar particles such as electrons interact to gain their all-important masses. Indirect evidence for the existence of the Z

boson was [obtained at CERN in 1973](#), and both the W and Z bosons were finally [discovered at CERN](#) a decade later, their masses being as predicted by the electroweak theory. The existence of the Brout-Englert-Higgs field remained to be proven and, to do that, the only hope was to detect the associated particle, known as the Higgs boson.

After the attempts of the experiments at the [Large Electron-Positron Collider](#) (LEP) at CERN and at the Tevatron at Fermilab in the United States, the boson hunters placed all their hope in the [Large Hadron Collider](#) (LHC), which has been generating higher-energy collisions since 2010. At the end of 2011, the two general-purpose LHC experiments, ATLAS and CMS, presented promising early results that were nonetheless still inconclusive. The LHC restarted in April 2012 at a slightly higher energy after a technical maintenance stop in the winter. Data quickly revealed the presence of a particle with properties that matched those of the long-sought Higgs boson, which culminated in the announcement of 4 July. One year later, the [Nobel Prize in Physics was awarded jointly to François Englert and Peter Higgs](#). The Nobel academy mentioned CERN and the ATLAS and CMS experiments in the statement accompanying the prize.

Since the discovery, the two experiments have carried out a lot of work to [define the new particle](#). For the Higgs boson is an exotic item in the particle zoo. As the only known elementary particle with zero “spin”, it could potentially shed light on profound open questions in fundamental physics – ranging from the decoupling of the electromagnetic and weak forces immediately after the Big Bang to the ultimate stability of the Universe. That’s why 4 July 2012 marked the start of a new adventure for particle physics.

Recollections

Fabiola Gianotti came to CERN as a research physicist in 1994. She was spokesperson of the ATLAS experiment at the Large Hadron Collider (LHC) from March 2009 to February 2013. On 4 July 2012, alongside the CMS spokesperson at the time, Joe Incandela, she announced the discovery of the Higgs boson. She has been CERN Director-General since 2016.

“It was 2012 and, as we usually do in our field to avoid biases, we had “blinded” a relatively small mass region that had not been excluded by previous searches for the Higgs boson. All of our analysis optimisation, Monte Carlo tuning, etc.,

was done outside this region. At the beginning of June 2012, we were ready to unblind our results for the Higgs to gamma–gamma channel.

I was at Fermilab in Chicago the day of the unblinding and I emailed one of the conveners, asking “Did you unblind? What did you find? Send me a plot.” He did and there it was, an excess at a mass of around 125 GeV, at exactly the same position as the hints from the 2011 data! I replied “my goodness”, to which he responded “indeed”. It was a very short exchange, just three words: we both knew that the Higgs boson was there.



On 4 July 2012, in front of a packed amphitheatre, Fabiola Gianotti, spokesperson for the ATLAS collaboration, presents the ATLAS results on the search for the Higgs boson. (Image: [CERN](#))

Results in only one channel weren't enough for a robust discovery claim. If it really was the Higgs boson, we had to see a handful of events in the four-lepton final state. We had nothing at the beginning of June. But all of a sudden, in the second half of June, candidates in the four-lepton channel began to pop up. The CMS collaboration had promising signals in the same mass region and, as spokespersons of the CMS and ATLAS collaborations, Joe Incandela and I shared our preliminary results with the then Director-General, Rolf Heuer, and began to prepare for an announcement. Joe and I were in contact daily. I knew what CMS had, he knew what ATLAS had, but we never disclosed this to our collaborations to avoid increasing the pressure and introducing any bias.

Those weeks in June 2012 were memorable: lots of emotion, lots of pressure, people working day and night, eating pizza at 3 a.m. at CERN. There was such an electricity, such an exciting atmosphere! The LHC was working fantastically well, delivering large amounts of data daily, and we had put in place fast-track procedures to calibrate and analyse data almost immediately. Despite the emotion and pressure, we were extremely focused, and we did zillions of checks and cross checks. It was an immense amount of work, done in a very short time. What really surprised me was that, despite the result at stake, no-one in the ATLAS or CMS collaborations – almost 6000 people – disclosed our findings. It was a strong sign of our responsibility, commitment and integrity.

On the morning of 4 July, I got to CERN and saw the crowds of people trying to enter the Main Auditorium. The room was so packed.

Joe Incandela spoke first. I remember that while he was showing his slides, I thought with gratitude of the thousands of colleagues that had worked on ATLAS, CMS and the LHC over the decades and all the people that had made the LHC and its experiments possible. Some of them were no longer with us to see such a great accomplishment.



CERN Director-General, Rolf Heuer (left), and ATLAS and CMS experiment spokespersons Fabiola Gianotti (centre) and Joe Incandela (right), following the announcement of the results on the search for the Higgs boson on 4 July 2012. (Image: [CERN](#))

When I started to speak, I looked at the audience and spotted a few ATLAS colleagues who became my reference points during the talk. They were looking at me with such intensity, and they really gave me strength and energy. The room exploded into a big round of applause when I announced the five-sigma significance. Straight after the presentations came the press conference. This all happened on a Wednesday, and Wednesday meant the LHC Machine Committee (LMC) meeting, which I always attended. So, I was there, of course, as on every Wednesday. I remember Mike Lamont, who was machine coordinator at the time, was very surprised to see me on that special day. He started his report with a slide that said: "Status report from the Higgs factory".

Finally, after a long and exhausting day, I went home and packed for the International Conference on High Energy Physics (ICHEP) in Melbourne. My flight was early in the morning on 5 July. At 3 a.m., I woke up suddenly, realising that Australia is on the other side of the globe and I'd packed for

summer not winter. I quickly repacked in the middle of the night! The next morning, I walked onto the plane, past the table of newspapers, and all the front pages showed the Higgs boson discovery. I thought, “oh my goodness, we are everywhere” and then fell fast asleep for most of the flight! The adrenaline had evaporated all of a sudden.

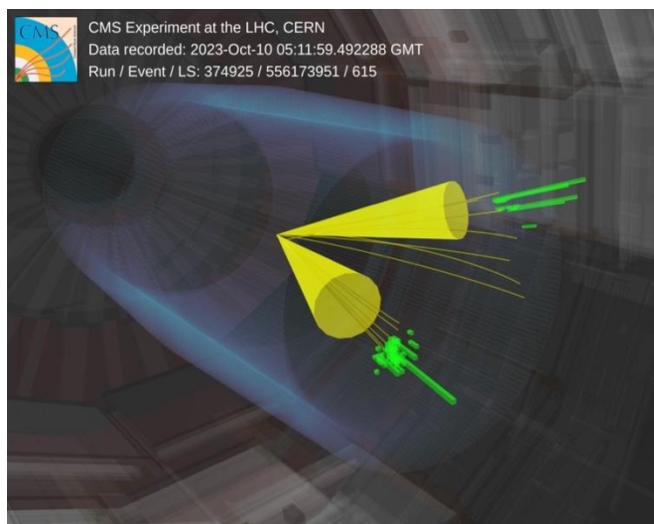
At ICHEP, everybody was excited, but very quickly we got back to business discussing the next steps. The 4 July announcement was just the beginning of the huge work that followed to understand a very special particle, which is still ongoing today.

It was an immense privilege to be the spokesperson of an experiment at the time of a monumental discovery. A discovery is teamwork, the result of decades of hard work by physicists, technicians, engineers and other personnel. I was representing this great community. I could feel that strength on 4 July 2012.

More information about the search for the Higgs boson and the research carried out since its discovery can be found in [the Higgs10 series](#) published in 2022, on the tenth anniversary of the discovery.

CMS uses photons to probe the structure of nuclei

Using data from the first heavy-ion run of LHC Run 3 in 2023, the experiment presents the first measurement of D^0 meson production in photon-lead collisions



Event display of a candidate for D^0 production at the CMS detector. (Image: CMS collaboration)

At the Large Hadron Collider ([LHC](#)), heavy ions are accelerated to extremely high energies, which creates strong electromagnetic fields. As a result, photons from the oncoming lead-ion beams can interact with each other or with the nuclei; these interactions are known as ultraperipheral collisions. Photon–nucleus scatterings at the highest energy that can be achieved with existing particle accelerators are useful probes that allow physicists to investigate the structure of nuclei. While the common picture of nucleons is that they contain three quarks (up–up–down for protons and up–down–down for neutrons), in reality, a complex sea consisting of quark–antiquark pairs and gluons makes up a large fraction of the proton

and neutron energies. Ultraperipheral collisions are an extraordinary tool to test the nature of nuclear matter.

The [CMS experiment](#) has recently released the first [results](#) using data from the first [heavy-ion](#) run of LHC Run 3. The results measure the production of D^0 mesons (containing a charm quark and an up antiquark) and their antiparticles, D^0 bar mesons (made of an up quark and a charm antiquark), in ultraperipheral collisions for the first time. D^0 mesons are formed by charm quarks that are kicked out of the nuclei by the photons and carry information about parton distribution functions, which describe how quarks and gluons behave inside nuclei.

To measure D^0 production, the CMS detector first selects events in which photon and lead nuclei collisions have caused the latter to break up. When this happens, neutrons flow from the collision in parallel to the beam, whereas protons and intact nuclei will follow a curved path as their charge interacts with the LHC’s magnetic fields. Two calorimeters, at zero degrees to the beam and located 140 m away on either side of the interaction point, are able to detect such neutrons. If they are seen in one calorimeter and not the other, in a time window consistent with the collision, this event is selected for further investigation.

Then, the products of the D^0 decay – oppositely charged kaon and pion pairs – are reconstructed in

the CMS detector. Physicists consider all combinations of pion and kaon trajectories, with each track taking an assumed mass of the kaon and pion. They then filter these combinations using the data to identify tracks that match what they expect from a D^0 meson. From this, they are able to measure the so-called production cross section, which is the rate at which D^0 mesons are produced.

For CMS, the study of nuclear structure using D^0 meson production is just one of many applications of ultraperipheral collisions. With

time, as methods are refined and systematic uncertainties are reduced, this technique will be able to constrain the parton distribution functions, allowing physicists to understand the structure of nuclear matter more deeply.

The next heavy-ion run of LHC Run 3 will start at the beginning of November.

Read more:

- [CMS briefing](#)
- [Paper](#)

CMS collaboration

Curious connections: Geneva's Natural History Museum comes to CERN

Come and discover the links between particle physics and the natural world at the Globe's first temporary exhibition



Visitors exploring the "Curious connections" exhibition at the Globe of Science and Innovation. (Image: CERN)

Where can you see a Large Hadron Collider magnet next to a piece of volcanic rock, a dinosaur bone next to a neutrino horn, the Standard Model with the tree of life or dinosaur tracks next to particle tracks?

From 26 October 2024 to 2 March 2025, the exhibition *Curious connections: the Museum comes to CERN*, for visitors aged eight and up, is revealing fascinating links between particle physics and natural history.

The result of a collaboration between CERN and the [Natural History Museum of Geneva](#), this is the

first temporary exhibition to be hosted in CERN's new exhibition space at the Globe of Science and Innovation. It showcases part of the Natural History Museum's collections during its closure for renovations, creating connections with objects from CERN and providing an intriguing new way to discover the two institutions.

Interested? [Become a guide!](#) The exhibition needs guides to show visitors around, represent CERN and help decode bubble tracks. The exhibition is open during CERN Science Gateway opening hours (Tuesday to Sunday, 9.00 a.m. to 5.00 p.m.).

Don't miss: *Danser avec l'Evolution*

Danser avec l'Evolution, a related event that blends contemporary dance with physics and life sciences, is coming to CERN Science Gateway on 19 November at 7.30 p.m. This dance performance tells the story of human history, drawing connections between evolution and the mysteries of the cosmos. After the performance, the paleoanthropologist Pascal Picq and the physicist Leïla Haegel will discuss the links between humankind and the Universe. The event will be in French and you can register for it via [Indico](#).

Charting the future of neutrino experiments

Different projects for next-generation neutrino experiments are under way at CERN to help to shed light on the nature of these elusive particles

To help to shed light on the nature of elusive neutrino particles, CERN's [Neutrino Platform](#) enables a global community of neutrino experts to develop and prototype different projects for next-generation neutrino experiments. From DUNE at LBNF in the United States to T2K in Japan, developments for far and [near detectors](#) are advancing well. Alongside this, novel identification techniques for neutrino beams are vital to strongly reduce systematic uncertainties on key observables – such as the flavour-dependent neutrino flux. These techniques are being developed in the framework of the [Physics Beyond Colliders study](#), in particular by the ENUBET and NuTAG collaborations.

ENUBET aims to exploit the fact that every time a neutrino is produced, it is accompanied by a charged lepton that can be detected and identified in a calorimeter with very good resolution. To detect these charged leptons, physicists envision a fully instrumented 40-metre-long decay tunnel.

The collaboration has just completed a first successful test of its demonstrator's full data acquisition system.

NuTAG proposes to follow the concept of a tagged neutrino beam to study neutrino oscillations using silicon detector technology. A proof-of-principle of this technique has been recently demonstrated at the NA62 experiment at CERN, where two tagged neutrino candidates have been found for the first time.

Combining these two concepts would result in a tagged neutrino beam with full particle identification. At the moment, a feasibility study is under way to find a suitable site for a possible first implementation.

This is a short summary of [this article](#) published in the newsletter of CERN's EP department

Kristiane Bernhard-Novotny

Computer Security: 2FA – 2 Fine Astuces

With the 2nd Fase Accomplished, the roll-out of 2-Factor Authentication (2FA) will soon be complete. After enrolling all CERN staff in 2023, in 2024 we deployed 2FA to about 12 700 computing accounts linked to CERN's user community (i.e. those fine folks who come to CERN about 5–100% a year). The final step is 2025's Full Activation of 2FA for all so-called CERN "participants" (i.e. about 7300 accounts of people who work with but never come to CERN). And, apart from some loose ends still to be tied up, we are done!

With this, CERN has put in place an essential pillar of protecting its computing facilities. 2 Fantastic Advantages to ensure that a single password cannot compromise important computing services, control systems or data storages. One of the ultimate silver bullets against [ransomware attacks](#).

To make your life under 2FA a bit easier, however, here are 2 Fine *Astuces* (tips) to improve 2FA usability:

- If you have more than one smart device, note that you can export/import your 2FA codes between smartphones. Just go to the "settings" menu of your favourite 2FA number generator (e.g. Aegis or Ente): "Export" the codes, transfer them to your other phone, and "Import" it there. However, those exports cannot necessarily be imported into other brands of 2FA number generator.
- If you prefer a "Yubikey" 2FA token but are fed up with the large format of your Yubikey, consider collecting a smaller one from the IT SOS Helpdesk in CERN's Restaurant 2. The Yubikey "nano" is perfect to be left in your laptop all the time* as it does not easily bend or break when the laptop is transported. But

consider ordering the USB-C version as the USB-A one activates too easily on touch.

- On the other hand, if you want to get rid of your Yubikey altogether, laptops with fingerprint readers allow you to configure those readers in the same way as your Yubikey (actually, the underlying FIDO2 protocol is the same). However, as the corresponding enrolment depends strongly on your laptop model, its operating system and the browser used, we can't give more precise instructions. But we know that many people have managed.
- (But, no!, using your password manager for 2FA is NOT an option and is [prohibited by CERN's Computing Rules](#)).
- And, as a bonus, if you want to try out 2FA when SSHing into LXPLUS/LXTUNNEL, check out [this pilot project](#), and note that ["ControlMaster" for multiplexing connections is your friend](#).

(OK, those are actually 2+2 Fine Astuces, but who cares?)

Finally, here are 2 Funny Anecdotes reported to us by users of 2FA: one user works for an institute

that neither allows smartphones on site nor has USB ports for using Yubikeys enabled. Hmmmm? But instead of bringing in a so-called ["Token2"](#) device (which itself might not be allowed), that user calls his partner over a landline connection to obtain the necessary 6-digit code. The 2nd Funny Anecdote (or not?) is the user who claimed burnout due to the psychological stress of always having the 2FA token with them and, thus, always being reachable by their partner... Fortunately, those 2 Funnies Are extremely rare among the 32k users of 2FA today!

P.S. Did you count how often 2FA appeared in this text? We count 28 Full Appearances.

**Yes, this means a little reduction in computer security, but at least you will notice if your laptop disappears and your 2FA is compromised. In that respect, the loss of your laptop/Yubikey combined is the equivalent of losing your smartphone.*

Computer Security Office

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

31 Oct 19:30 | Knowledge Sharing | CERN Science Gateway | Public event | FR (simultaneous interpreting in EN)

[Dark Matter Day 2024](#)

4–15 Nov | Knowledge Sharing | Main Building Exhibition | CERN & Society Foundation | EN

[CERN & Society Foundation: 10 years bridging science and society](#)

4 Nov 13:00 | Knowledge Sharing | CERN & online | CERN Knowledge Transfer group | EN

[CERN Venture Connect Summit 2024](#)

6 Nov 10:00 | At CERN | Online | Human Resources | EN

[MERIT 2025: Public Meeting in English](#)

6 Nov 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[Meet the author of « Le CERN, quelle aventure! » \("Your adventures at CERN"\)](#)

8 Nov 9:00 | At CERN | Main Auditorium & online | HSE unit | EN

[Environment Town Hall](#)

8 Nov 11:00 | Knowledge Sharing | The Globe of Science and Innovation | IdeaSquare | EN

[IdeaSquare 10th anniversary](#)

8 Nov 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 13 – Timeboxing](#)

8 Nov 19:30 | Knowledge Sharing | CERN Science Gateway | Public event | EN (simultaneous interpreting in FR)

[Global Challenges and the role of science, an evening with Al Gore](#)

11–15 Nov | Physics | 4/3-006 | TH Institute | EN
[Light ion collisions at the LHC bringing together theorists, experimentalists and accelerator physicists](#)

14 Nov | Knowledge Sharing | CERN | Education locale | FR
[Futur en tous genres](#)

16 Nov 8:30 | Knowledge Sharing | UniMail, Geneva | Education locale | FR
[Élargis tes Horizons: Journée ateliers scientifiques et technologiques pour les jeunes filles](#)

18 Nov 14:00 | At CERN | Online | Human Resources | FR
[MERIT 2025: Réunion publique en français](#)

19 Nov 19.30 | At CERN | CERN Science Gateway | Public event | FR
[Danser avec l'Évolution](#)

Registration now open

26 Nov 13:00 | Knowledge Sharing | Online | KT Seminars | EN
[CERN Workshop on Global Health: Presentation of Results Webinar](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
[DRD1 Gaseous Detectors School](#)

6 Dec 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[Meet the authors of "The stars' last dance" \("La dernière danse des étoiles"\)](#)

11 Dec 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
[Meet the authors of « Les Aventures Extraordinaires des Globifiques »](#)

10–14 Mar | Physics | Ferney-Voltaire | CERN Accelerator School | EN
[Basics of Accelerator Physics and Technology](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Ombud's corner

Proactively tackling conflicts

Conflict resolution in the workplace takes time, good listening skills and a willingness to find solutions

Taking the time to listen to all parties concerned and to hear everyone's point of view allows the mediator to better understand the situation. Active listening allows the different parties to tell their side of the story and to explain how the events have affected them, what they need and how they wish to move forward.

Emotions will inevitably rise to the surface and it is important to acknowledge them without judgement. When difficult situations are brought to my attention, I sometimes simply say "I can hear anger in your voice". Negative emotions are often the sign of a deep need that isn't being satisfied or of a feeling of loss.

The next step is to invite the parties to set out, clearly and precisely, what they require to have

their needs met, enabling them to take back a level of control over the situation that is bothering them.

In conflicts between two colleagues, I have observed in each of them similar needs for respect and for mutual trust. It is therefore vital to facilitate discussions that may allow them to find practical solutions that are satisfactory in the long term; the parties will then be able to work reasonably well together.

Proactively tackling conflicts often involves calling on a trusted third party, either in the workplace or through the support structures that exist at CERN, of which the Ombud is just one.

If you would like to talk more about this subject or its application in a particular situation, please

come and see me. As Ombud, my main purpose is to provide an unofficial channel for conflict resolution; my role is also to offer my opinion on the implementation of the Code of Conduct.

Thank you for your attention, and I look forward to seeing you soon.

Marie-Luce Falipou

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>.

The Ombud is available from Monday to Friday in office 500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at ombuds@cern.ch.

More information can be found on the Ombud's website: <https://ombuds.web.cern.ch/>
